Lab Assignment-3

Calculations:-

Calculations for Two plays:-

The incremental Sample-average formula is:

$$Q_{new} = Q_{old} + \frac{R - Q_{old}}{N}$$

Play 1:

Initially:

$$Q(B) = 0, \quad N(B) = 0$$

Suppose machine B is selected and recieves reward 6.

$$N(B) = 1$$

$$Q(B) = 0 + \frac{6-0}{1} = 6$$

Therefore:

$$N(B) = 1$$

$$R(B) = 6$$

$$Q(B) = 6.00$$

Play 2:

Suppose machine B is selected again and recieves reward 7.

$$N(B) = 2$$

$$Q(B) = 6 + \frac{7-6}{2}$$

$$Q(B) = 6 + 0.5 = 6.5$$

Therefore:

$$N(B) = 2$$
$$R(B) = 13$$
$$Q(B) = 6.50$$

Total Reward received:

The total reward is calculated as:

total_reward = sum(R).